

DIGITAL IC DESIGN

SA-8: Serial 8-bit Adder

1. Aum Bavarva — 202404004
2. Preet Patel — 202404027
3. Om Hirapara — 202404011



Dhirubhai Ambani University
Formerly DA-IICT

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1 Designs

1.1 D Flip-Flop

Schematic:

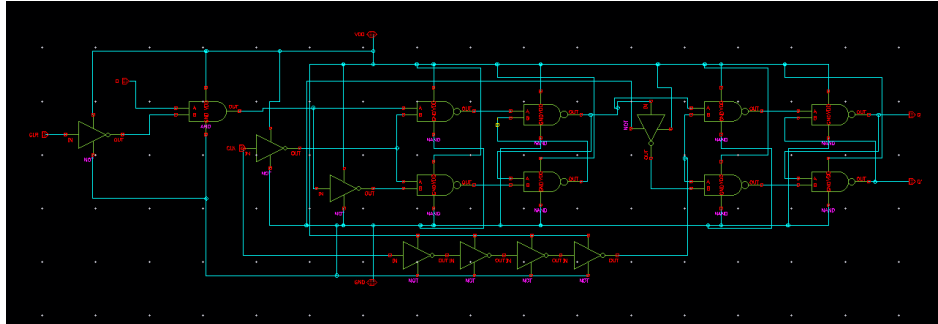


Figure 1: D Flip-Flop Schematic

Symbol:

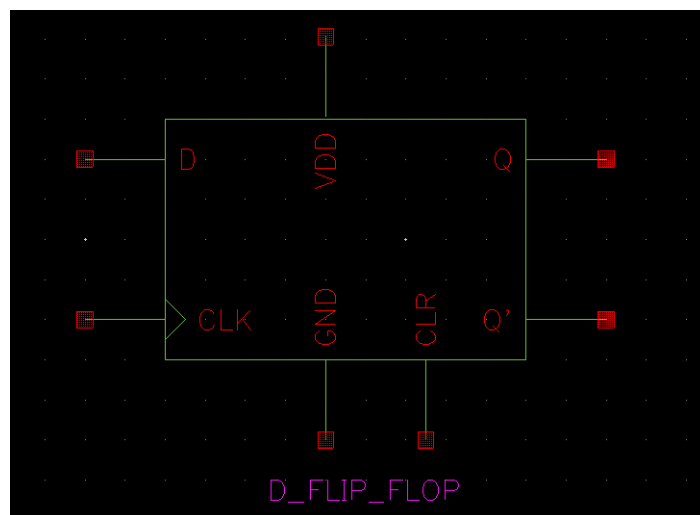


Figure 2: D Flip-Flop Symbol

1.2 MUX 2×1

Schematic:

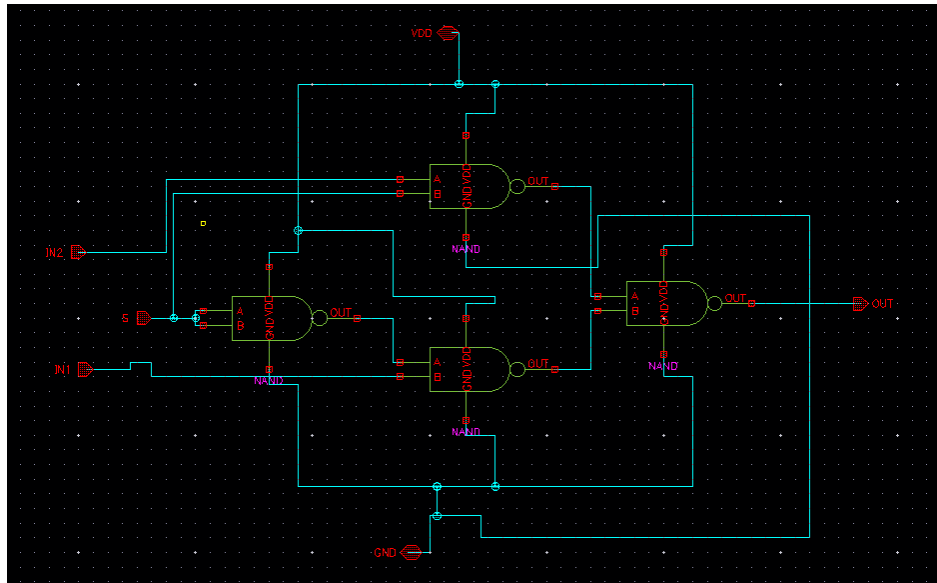


Figure 3: MUX 2×1 Schematic

Symbol:

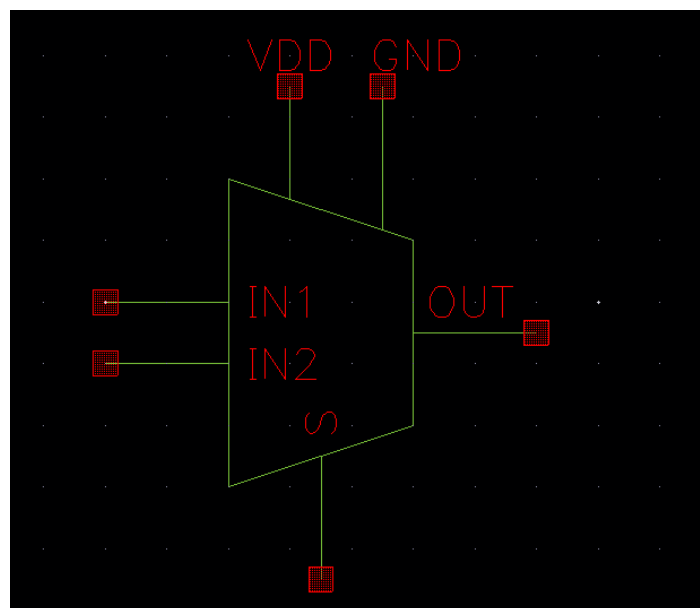


Figure 4: MUX 2×1 Symbol

1.3 OR Gate

Schematic:

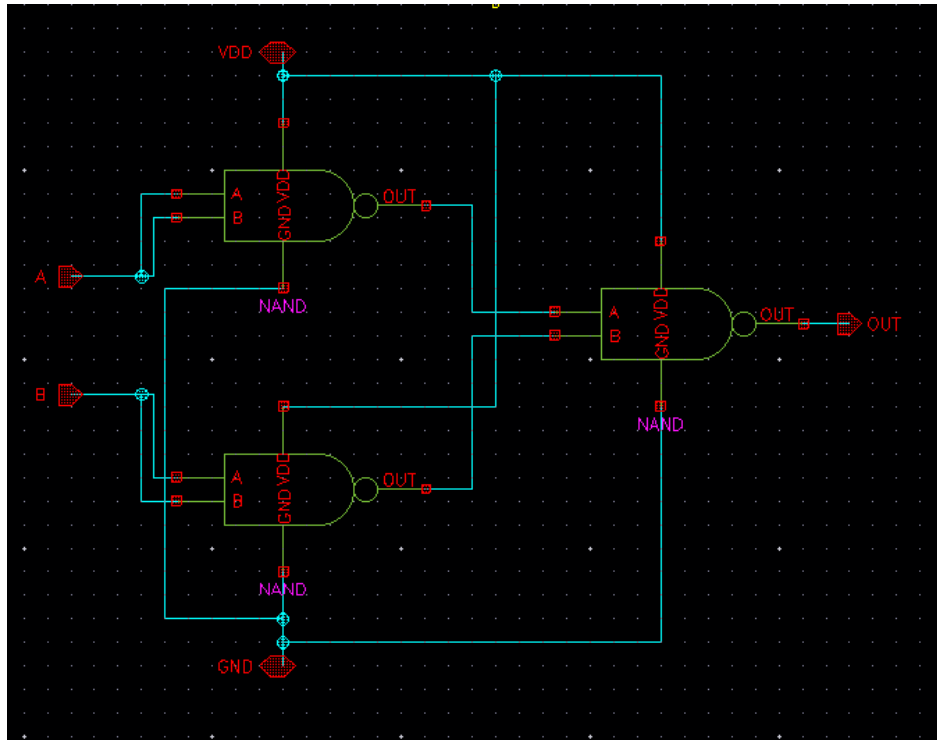


Figure 5: OR Gate Schematic

Symbol:

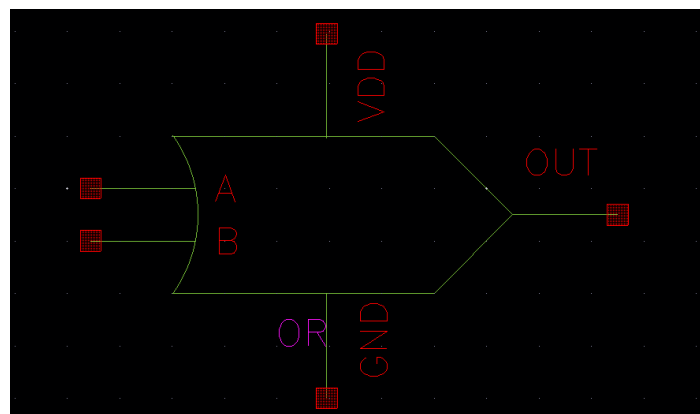


Figure 6: OR Gate Symbol

1.4 1-bit Full Adder

Schematic:

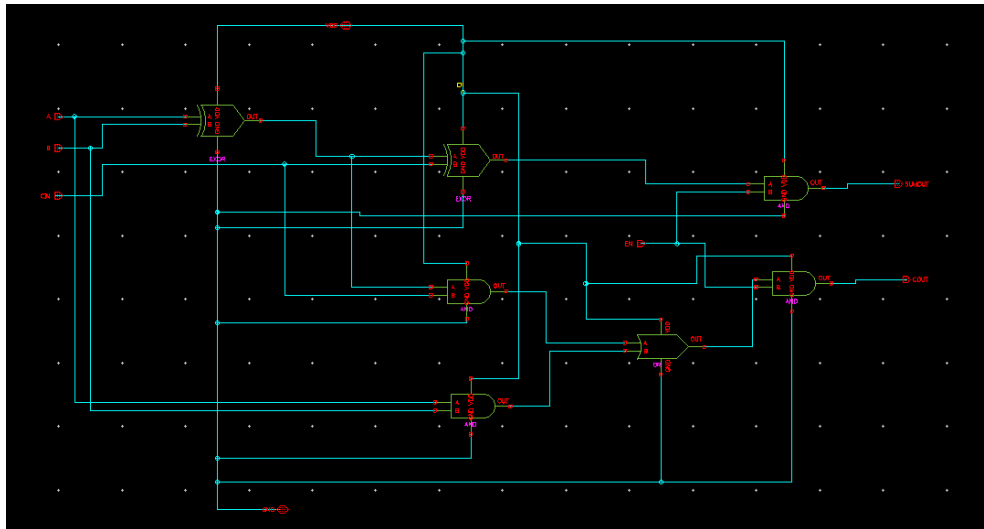


Figure 7: 1-bit Full Adder Schematic

Symbol:

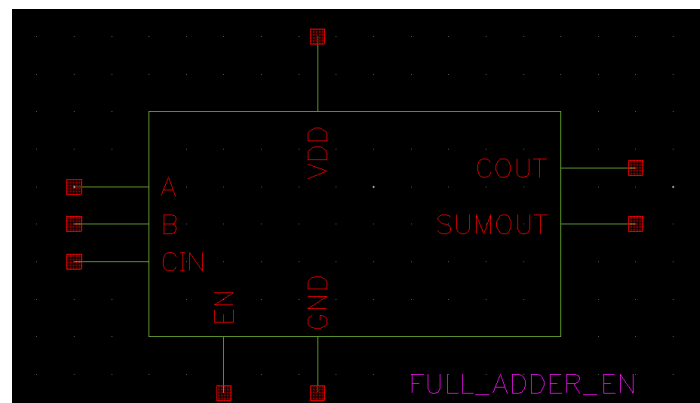


Figure 8: 1-bit Full Adder Symbol

1.5 17-bit Input SISO Shift Register

Schematic:

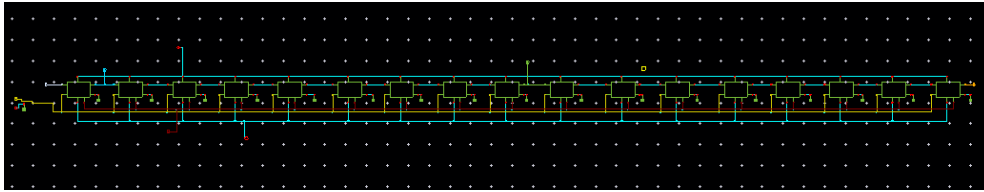


Figure 9: 17-bit Input SISO Schematic

Symbol:

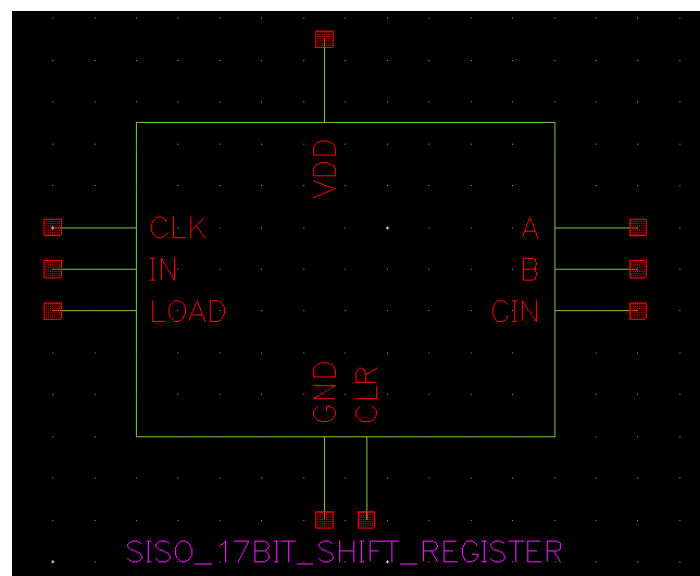


Figure 10: 17-bit Input SISO Symbol

1.6 9-bit Output SISO Shift Register

Schematic:

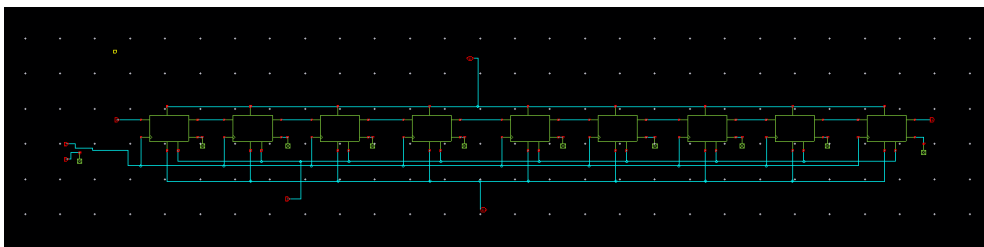


Figure 11: 9-bit Output SISO Schematic

Symbol:

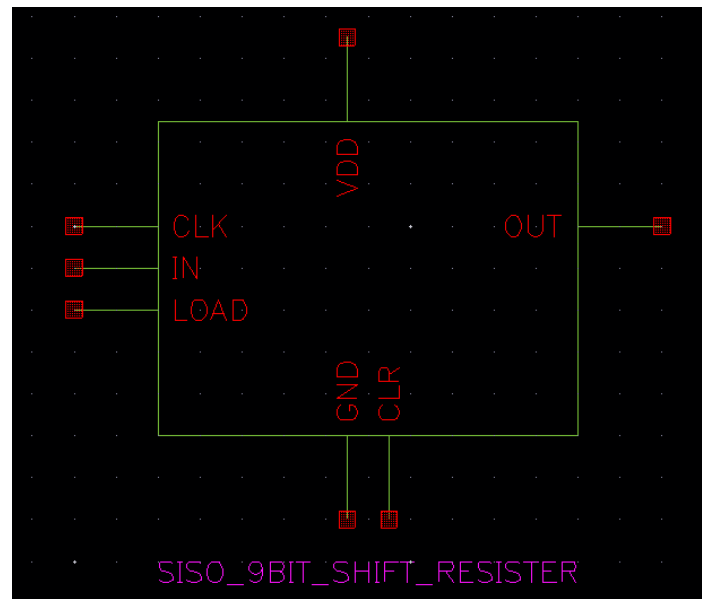


Figure 12: 9-bit Output SISO Symbol

1.7 8-bit Serial Adder

Schematic:

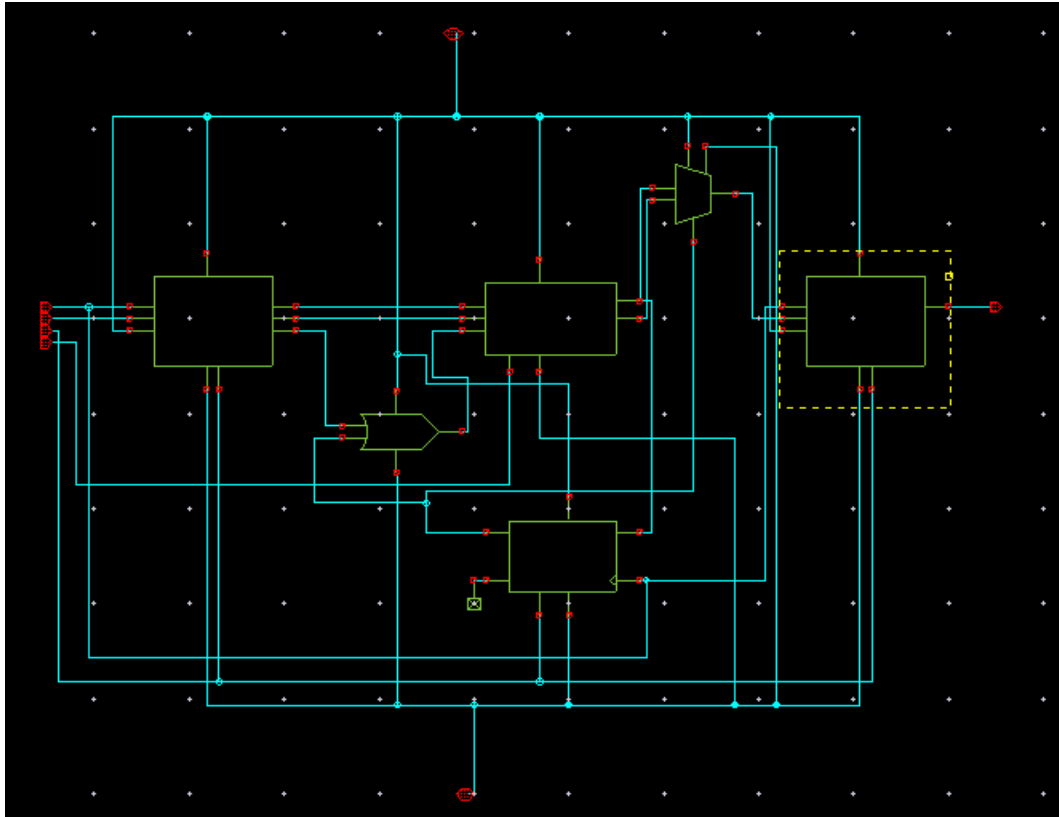


Figure 13: 8-bit Serial Adder Schematic

Symbol:

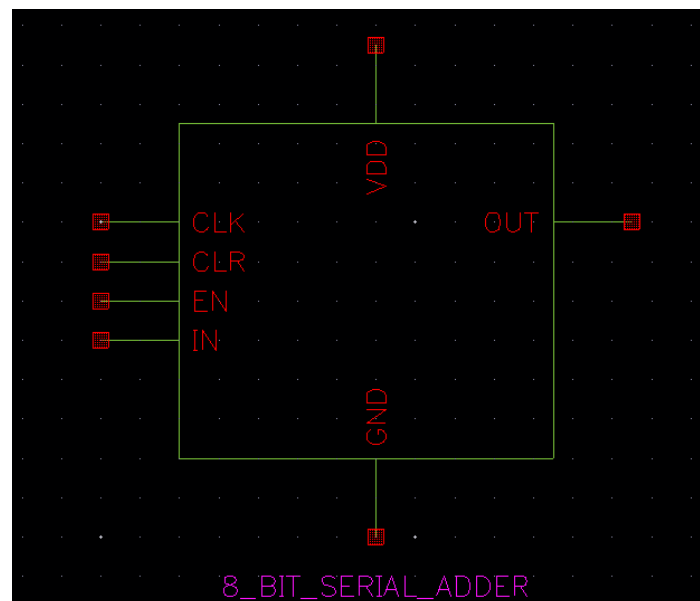


Figure 14: 8-bit Serial Adder Symbol

2 Description of Modules

2.1 D Flip-Flop

A standard positive edge-triggered D flip-flop. It is the fundamental storage element used throughout the design — most critically as the **carry register (R3)** that feeds C_{OUT} back as C_{IN} on each subsequent addition cycle. It has two control pins: **VDD** and **GND**.

2.2 MUX 2×1

A 2-to-1 multiplexer placed between the full adder outputs and the 9-bit output SISO. Its select line is driven by the **EN** signal:

- $EN = 1 \rightarrow$ passes **SUM** to the output register (addition phase)
- $EN = 0 \rightarrow$ passes **C_{OUT}** to the output register (carry capture phase)

2.3 OR Gate

Used to generate C_{IN} for the full adder. After all 17 input bits are loaded into the input SISO, the serial IN line is guaranteed to be 0. The OR gate therefore passes the stored carry-out cleanly as the carry-in for the next cycle without any spurious input.

2.4 1-bit Full Adder

The sole combinational computation block. It accepts one bit of A, one bit of B, and a C_{IN} , and produces a SUM bit and C_{OUT} .

- **Inputs:** A, B, C_{in}
- **Outputs:** SUM, C_{OUT}
- **Worst-case propagation delay:** $t_{pd} \approx 0.51$ ns

2.5 17-bit Input SISO Shift Register

A 17-stage serial-in serial-out shift register (R1) for loading the two operands and carry-in. Data is clocked in serially over 17 cycles and clocked out one bit per cycle to the full adder during the addition phase.

2.6 9-bit Output SISO Shift Register

A 9-stage SISO register (R2) for capturing the result. It stores 8 SUM bits (S0–S7) and 1 C_{OUT} bit, then shifts the complete 9-bit result out serially on the OUT pin.

2.7 8-bit Serial Adder (Top-Level)

The top-level module integrating all sub-blocks. It has 7 pins:

Pin	Direction	Description
IN	Input	Serial data input
CLK	Input	Clock (positive edge-triggered)
CLR	Input	Asynchronous clear (active high)
EN	Input	Enable — controls addition phase and MUX
OUT	Output	Serial data output
VDD	Power	Supply voltage
GND	Power	Ground

3 Control Pin Usage

CLK: Drives all flip-flops on the positive edge. Minimum clock period: $T_{min} = 1.41535$ ns, giving a maximum operating frequency of **714.2 MHz**.

CLR: Asynchronous clear. When $CLR = 1$, all registers reset to 0 immediately, independent of CLK.

EN: The primary control signal governing the three operational phases:

- **EN = 0 (Input Phase):** Data shifts into R1; MUX routes C_{OUT} to R2.
- **EN = 1 (Addition Phase):** Full adder operates; MUX routes SUM to R2.

EN transition sequence:

1. Set **EN = 1** at the negative edge of clock cycle 17 (all inputs loaded).
2. Set **EN = 0** at the negative edge of clock cycle 25 (all SUM bits registered, C_{OUT} latched in D-FF).
3. On the positive edge of cycle 26, MUX now selects $C_{OUT} \rightarrow$ carry is registered as the 9th output bit.

4 Functional Analysis of Sub-Modules

4.1 D Flip-Flop

Timing parameters were extracted from the transient simulation of the carry D flip-flop (R3).

Table 1: D Flip-Flop Measured Timing Parameters

Parameter	Symbol	Measured Value
Hold Time	t_{hold}	0.2116 ns
Setup Time	t_{su}	0.2998 ns
Clock-to-Q	t_{c-q}	0.3399 ns

4.2 1-bit Full Adder

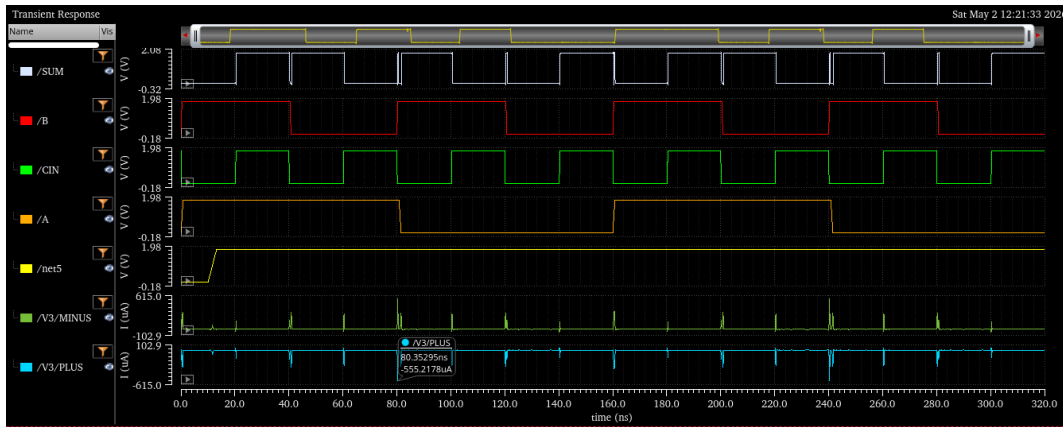


Figure 15: 1-bit Full Adder Simulation Waveform

Average propagation delays:

$$(t_{SUM})_{avg} = \frac{t_{[A,B]} + t_{cin}}{2} = \frac{0.4120 + 0.2523}{2} = 0.33215 \text{ ns}$$

$$(t_{COUT})_{avg} = \frac{t_{[A,B]} + t_{cin}}{2} = \frac{0.395745 + 0.33095}{2} = 0.36439 \text{ ns}$$

Table 2: Full Adder Worst-Case Propagation Delays

Output	Max. Delay
SUM	0.5025 ns
C _{OUT}	0.4812 ns
Worst-case t_{pd}	$\approx 0.51 \text{ ns}$

4.3 17-bit Input SISO Shift Register

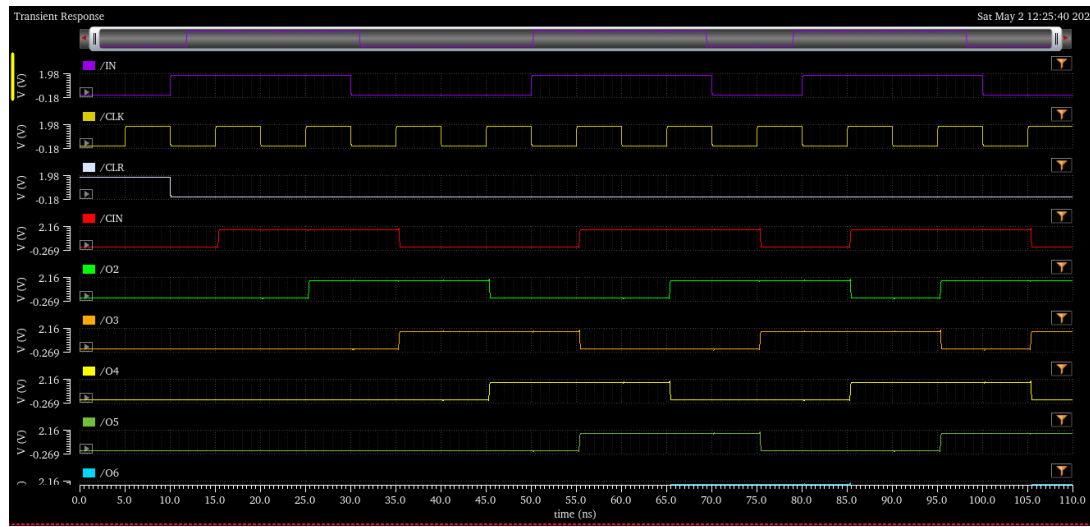


Figure 16: 17-bit Input SISO Simulation — Correct serial loading of A[7:0], B[7:0], C_{in} over 17 clock cycles

5 8-bit Serial Adder: Full Analysis

5.1 Block Diagram

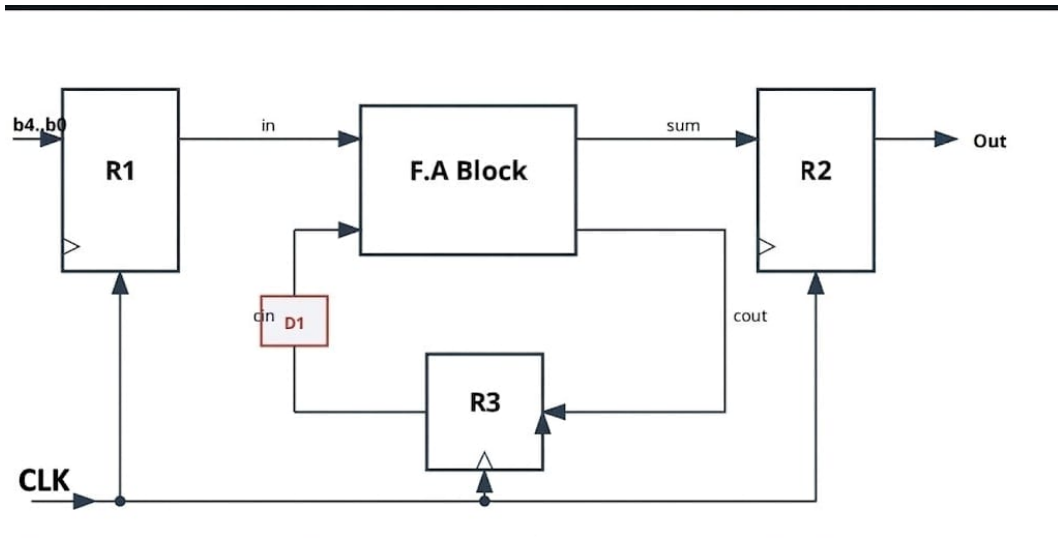


Figure 17: SA-8 Internal Block Diagram

The block diagram shows the three principal registers — R1 (17-bit input SISO), R2 (9-bit output SISO), and R3 (carry D-FF) — connected via the 1-bit full adder and the 2×1 MUX.

5.2 Workflow

The SA-8 completes one addition in **35 clock cycles** across three phases.

Table 3: Clock Cycle Breakdown

Phase	Description	Cycles	Count
1	Input loading into R1	1 – 17	17
2	Addition, SUM bits into R2	18 – 25	8
3	Carry capture into R2	26	1
4	9-bit result out on OUT	27 – 35	9
Total			35

Phase 1 — Input Loading (Cycles 1–17):

- Data clocked into R1 serially in order: C_{in} , $B[0]–B[7]$, $A[0]–A[7]$
- $EN = 0$; $CLR = 0$
- At the *negative edge* of cycle 17 → EN is set HIGH

Phase 2 — Addition (Cycles 18–25):

- $EN = 1$; full adder operates one bit per cycle, LSB first

- MUX selects SUM $\rightarrow$ each sum bit registered into R2
- At cycle 25 (8th addition cycle): final SUM bit registered in R2; C_{OUT} latched in carry D-FF
- At the *negative edge* of cycle 25 $\rightarrow$ EN is set LOW

Phase 3 — Carry Capture (Cycle 26):

- $EN = 0 \rightarrow$ MUX now selects C_{OUT}
- On the positive edge of cycle 26, C_{OUT} is registered into R2 as the 9th bit, completing the result

Phase 4 — Serial Output (Cycles 27–35):

- All 9 bits (S[0]–S[7], C_{OUT}) shift out serially on the OUT pin over 9 clock cycles

5.3 Testing

Testbench Schematic:

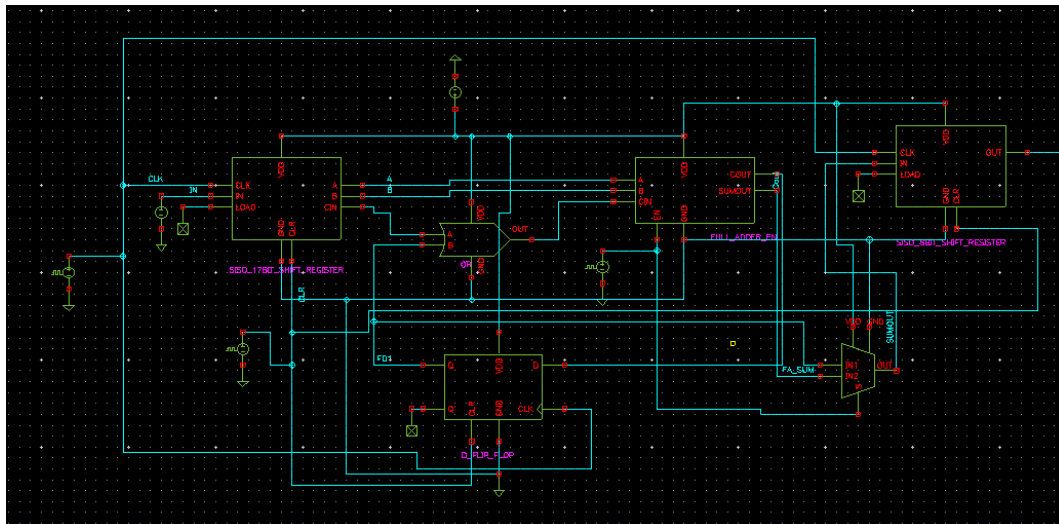


Figure 18: 8-bit Serial Adder Testbench Schematic

Overall Simulation Result:

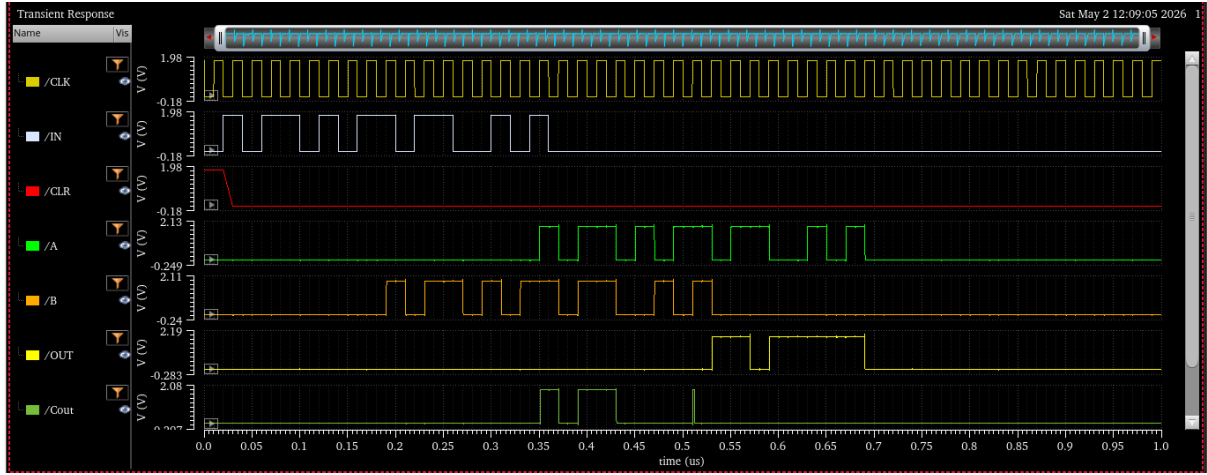


Figure 19: Simulation Waveform — Full 35-cycle Operation

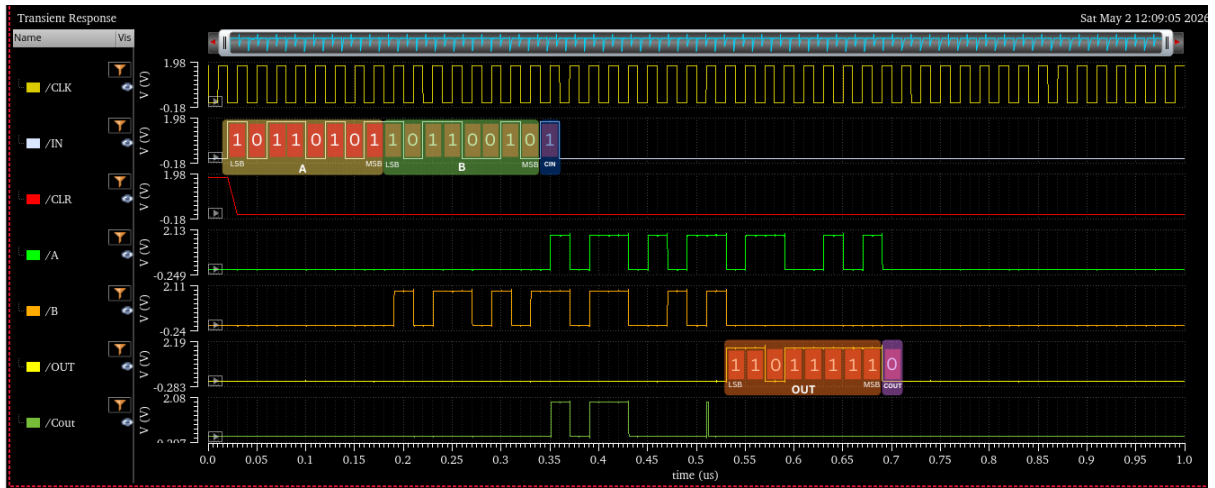


Figure 20: Simulation Waveform — Input and Output Highlighted

5.4 Analysis

5.4.1 Timing

The minimum clock period must satisfy the following constraint, accounting for all sequential and combinational delays in a single cycle:

$$T \geq t_{su} + t_{c-q} + t_{logic} + t_{pd}$$

$$T = 0.2998 + 0.3399 + 0.51 + 0.2664 = \mathbf{1.41535 \text{ ns}}$$

Hold-time verification:

$$t_{hold} < t_{edlogic} + t_{cdf} \implies 0.2116 < 0.205 + 0.3399 = 0.5449 \text{ ns} \quad \checkmark$$

Table 4: Timing Summary

Parameter	Symbol	Value
Setup Time	t_{su}	0.2998 ns
Clock-to-Q Delay	t_{c-q}	0.3399 ns
FA Worst-Case Delay	t_{pd}	0.51 ns
Combinational Logic Delay	t_{logic}	0.2664 ns
Hold Time	t_{hold}	0.2116 ns
Minimum Clock Period	T_{min}	1.41535 ns
Maximum Operating Frequency	f_{max}	714.2 MHz

5.4.2 Current

Current was measured under three input conditions to characterise switching behaviour.

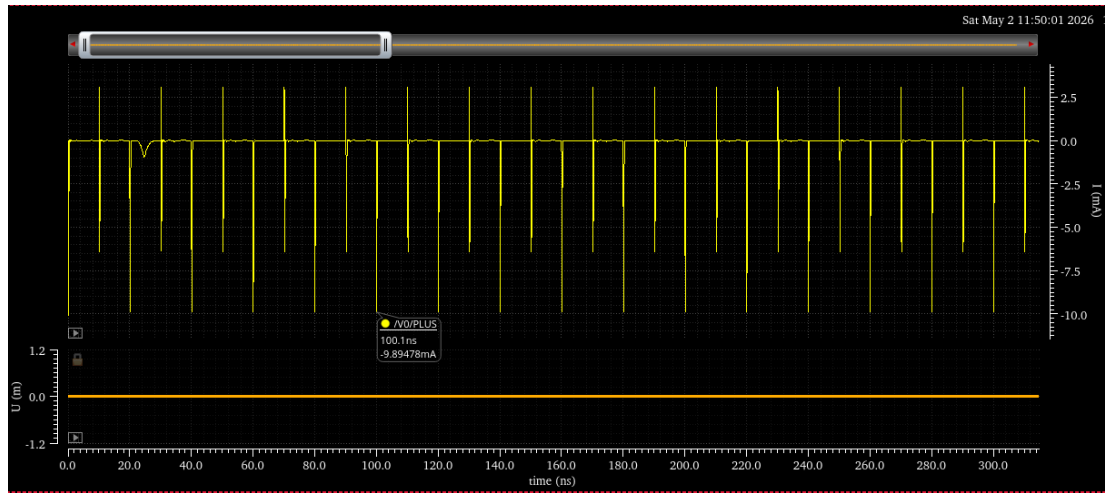


Figure 21: Current Waveform — All Inputs = 0 ($P_{avg} = 237.4 \mu\text{W}$)

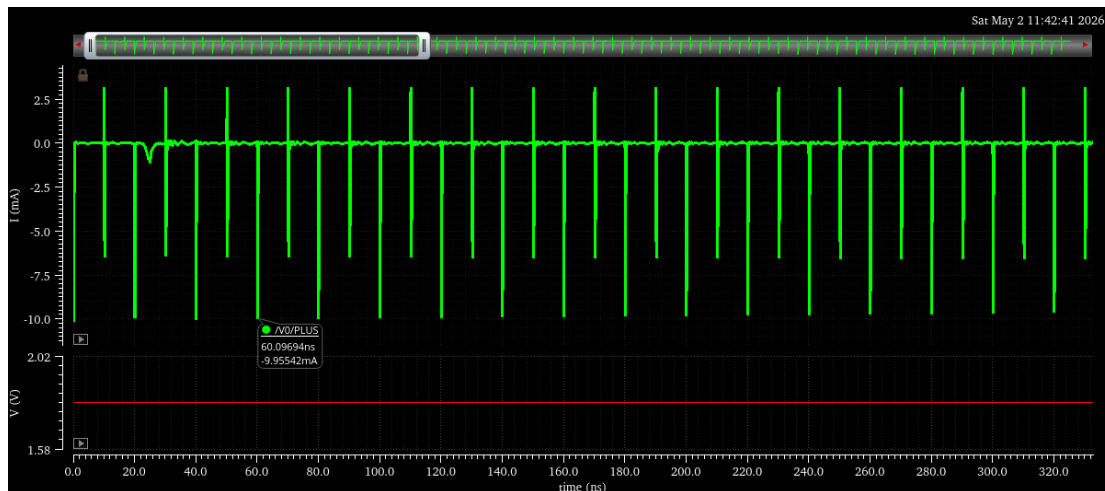
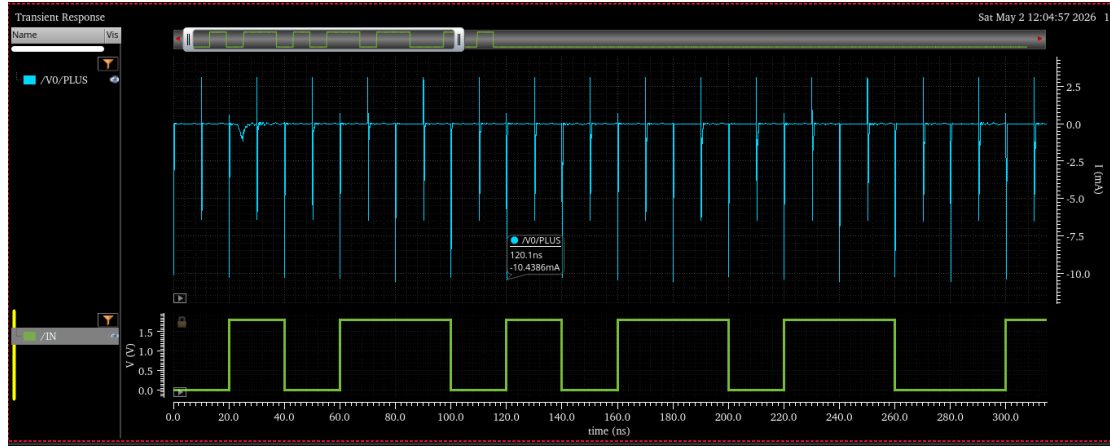


Figure 22: Current Waveform — All Inputs = 1 ($P_{avg} = 244.7 \mu\text{W}$)

Figure 23: Current Waveform — Random Input ($P_{avg} = 274.6 \mu\text{W}$)

5.4.3 Power & Energy

Power was analysed at two operating speeds to characterise the design across a range of clock rates.

Table 5: Power Dissipation Analysis

Parameter	Symbol	Case 1 ($T = 4 \text{ ns}$)	Case 2 ($T = 20 \text{ ns}$)
Peak Power	P_{max}	67.15 mW	18.67 mW
Average Power	P_{avg}	3.783 mW	0.264 mW
Energy/Op.	E_{avg}	$\leq 0.18530 \text{ nJ}$	$\leq 0.18522 \text{ nJ}$

The energy per operation remains constant at $E_{avg} \approx 0.1852 \text{ nJ}$ across both cases, confirming CMOS power-proportionality behaviour: as frequency scales down, average power reduces proportionally, leaving energy per operation invariant.

5.4.4 PDP and EDP

The Power-Delay Product (PDP) quantifies energy per switching event; the Energy-Delay Product (EDP) additionally penalises circuits with longer delays.

Table 6: PDP and EDP Summary

Metric	Formula	Case 1 ($T = 4 \text{ ns}$)	Case 2 ($T = 20 \text{ ns}$)
P_{avg}	—	3.783 mW	0.264 mW
t_{pd}	—	0.51 ns	0.51 ns
PDP	$P_{avg} \times t_{pd}$	$\sim 1.93 \text{ fJ}$	$\sim 0.135 \text{ fJ}$
EDP	$P_{avg} \times t_{pd}^2$	$\sim 0.98 \text{ fJ}\cdot\text{ns}$	$\sim 0.069 \text{ fJ}\cdot\text{ns}$
E_{avg}	$P_{avg} \times T_{process}$	$\leq 0.18530 \text{ nJ}$	$\leq 0.18522 \text{ nJ}$

6 Conclusion

The SA-8 serial 8-bit adder successfully computes the sum of two 8-bit operands in **35 clock cycles**, outputting a 9-bit result (8 sum bits + carry-out) serially. Key results are summarised below:

Figure of Merit	Value
Total Clock Cycles	35
Minimum Clock Period (T_{min})	1.41535 ns
Maximum Frequency (f_{max})	714.2 MHz
FA Worst-Case Delay (t_{pd})	0.51 ns
Hold Time Margin	0.2116 ns < 0.5449 ns ✓
Energy per Addition (E_{avg})	$\approx$ 0.185 nJ

The hold-time constraint is satisfied with adequate margin. The energy per addition operation remains consistent across both high-speed and standard operating modes, validating CMOS power-proportionality. The design is fully verified through Cadence Virtuoso simulation.